

Jack Birkin, Ph.D.

AI Engineer

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Professional Experience

AI Engineer

Famelia | Remote | Feb 2026 – Present

- As the sole AI Engineer, architected core LLM components within a production, Azure-hosted Python/LangGraph-based agentic pipeline, generating personalized recipe substitutions for families with heterogeneous dietary restrictions.
- Diagnosed a production latency problem by instrumenting the request path and analyzing traced data, identifying that 68% of median request time sat in a single sequential stage, then parallelized it for a measured 2.84x speedup (179s to 63s across a fixed five-recipe benchmark).
- Built a retrieval-augmented generation system end-to-end over a 3,500-page clinical nutrition reference: table-aware PDF ingestion and outline-based chunking into 1,844 chunks, Azure OpenAI embeddings, and pgvector/HNSW search in Supabase, serving a dietitian chatbot as grounded retrieval rather than prompt-stuffing.
- Introduced production engineering practice as the sole AI engineer: pytest gates on every deploy workflow (342 hermetic tests), end-to-end LangSmith tracing, and an offline evaluation harness replaying captured production cases to diff graded scores across commits.
- Implemented governance guardrails on LLM output, including allergen/restriction compliance scoring, automated validation against medical conditions, and human-in-the-loop dietitian approval before patient delivery.

Postdoctoral Research Associate

Texas A&M University, College Station, TX | Nov 2022 – Feb 2026

- Applied MCMC-based Bayesian modeling to noisy observational data to extract robust source parameters, supporting proposals totaling 800+ observation hours and contributing to 25 peer-reviewed publications (900+ citations).
- Constructed a Python pipeline aggregating low-SNR data across many sources, improving weak-feature detection 4x via four transformation methods, with bootstrap resampling and BIC model selection to guard against overfitting.
- Developed [cubespecfit](#), an open-source Python package for joint modeling of 3-D spatial-spectral data, and a Dash/Python [web application](#) on AWS for collaborative team-wide data review.
- Provided leadership within an international scientific collaboration, hosting team meetings and driving progress on key science goals, and served as postdoctoral mentoring lead in my department.

Graduate Researcher

Durham University, UK | Oct 2018 – Aug 2022

- Led cleaning and analysis of a 2 TB dataset, applying PCA-based background subtraction and stacking to recover faint signals.

Skills

Software: Python, SQL, LangGraph, Git, CI/CD, FastAPI, TypeScript, React, AWS, Azure, Linux, pandas, Dash

AI / LLMs: RAG, agentic AI, prompt engineering, multi-model pipeline design, LLM orchestration, Azure OpenAI, knowledge graphs, Claude Code

Data Science / Machine Learning: Regression, spectral data modeling, predictive modeling, data transformation, feature engineering, anomaly detection, model selection (BIC/AIC), uncertainty quantification, bootstrap resampling, MCMC

Leadership: Problem solving, project design and leadership, mentoring, presenting and communicating technical concepts

Education

Ph.D., Physics | Durham University, UK | Oct 2018 – Aug 2022

MPhys, Physics & Astronomy (First Class Honours) | Durham University, UK | Oct 2014 – Jun 2018

Certifications

AWS Certified Solutions Architect - Associate | Amazon Web Services

AWS Certified Cloud Practitioner | Amazon Web Services

Machine Learning Specialization | Stanford University & DeepLearning.AI